

When Visual Token Pruning Fails the Frontier: A Full-Matrix Diagnostic for Multimodal LLM Inference

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Abstract

Visual-token pruning can reduce multimodal LLM prefill cost, but a lower retained-token ratio is not enough to show that an inference policy preserves answer evidence or remains competitive with simple matched-budget controls. This distinction matters because many pruning policies look efficient in token count while silently changing the visual evidence available to the language model. We study this evaluation gap with AP-HTS, a diagnostic policy that combines spatial scaffolding, activation-based token scoring, and merge/prune variants, using Qwen3-VL-8B-Instruct on GQA, MMBench, TextVQA, and VQAv2. The full matrix contains 18,432 scored rows and 18,432 profile rows across full tokens, uniform pruning, FitPrune, AP-HTS, hybrid merge, and ablations, so each claim is tied to both accuracy and measured or analytical cost evidence. At the target 0.5 retained-token ratio, AP-HTS hybrid merge drops 0.079102 average accuracy relative to full tokens, whereas FitPrune drops only 0.009180 under the same protocol; AP-HTS does reduce analytical prefill FLOPs, but the accuracy loss concentrates on TextVQA, where OCR-sensitive evidence is removed despite an apparently reasonable token budget. The study therefore contributes a calibrated negative result: FitPrune is the current floor, TextVQA is the repair surface, and future MLLM token-reduction claims should report matched accuracy, retained-token ratio, and measured cost evidence together on the same evaluated backend.

1 Introduction

Multimodal large language models convert visual inputs into long sequences of patch, region, or projector tokens before the language model consumes them. Vision Transformers make this sequence length an explicit cost driver, and modern vision-language systems inherit that pressure when they feed visual embeddings into a decoder (Dosovit-

skiy et al., 2020; Liu et al., 2023a; Bai et al., 2023). Instruction-tuned and retrieval-style multimodal models further raise the stakes because their visual evidence can include small objects, written text, or layout cues that are easy to erase with a coarse token budget (Alayrac et al., 2022; Li et al., 2023; Dai et al., 2023). Recent token-reduction work therefore asks whether redundant image tokens can be pruned, merged, or packed without retraining the full model (Chen et al., 2024; Shang et al., 2024; Ye et al., 2024; Yang et al., 2024; Zhang et al., 2024; Wen et al., 2025). Benchmarks such as GQA, TextVQA, MMBench, and VQAv2 expose why the answer cannot be judged from token count alone: a policy can preserve general visual reasoning while losing OCR-critical evidence or vice versa (Hudson and Manning, 2019; Singh et al., 2019; Liu et al., 2023b; Goyal et al., 2017). This study evaluates that premise under a matched-budget full matrix. The gap is not whether pruning can reduce a token counter; many methods can. The harder question is whether a policy still preserves answer evidence when compared with simple and prior baselines at the same retained-token ratio, on benchmarks that include text-heavy and general visual reasoning cases. AP-HTS is useful for this question because its design combines three plausible ideas—spatial coverage, activation salience, and merge/prune recovery—that should have helped if the frontier were mainly a coverage problem.

The completed evidence rejects the intended positive frontier. AP-HTS was designed to combine a coarse spatial scaffold with activation-sensitive token allocation, then test whether hybrid merge/prune variants improve the accuracy-efficiency frontier. The four-benchmark matrix does show that pruning changes the expensive prefill path, yet it does not show a positive AP-HTS frontier. At the headline 0.5 retained-token ratio, AP-HTS hybrid merge averages 0.692187 accu-

racy, a 0.079102 drop from full tokens; FitPrune averages 0.762109, only 0.009180 below full tokens. Uniform pruning also exceeds AP-HTS at the same ratio. The correct conclusion is therefore not that AP-HTS succeeds, but that the current allocation rule is falsified by a stronger floor and by TextVQA failures. That distinction matters for an EMNLP/ACL paper package because a token-efficiency claim has three separable parts. The first part is the mechanism: which tokens are removed or merged, and whether the rule can be implemented inside an inference backend. The second part is the measurement: whether the expensive path actually sees fewer visual tokens, rather than only a bookkeeping counter after the model has already paid the cost. The third part is the frontier: whether the accuracy retained at a given token budget is competitive with simple and prior baselines. AP-HTS satisfies only the middle part in the current evidence. It changes the prefill FLOP calculation, but it fails the frontier and therefore cannot be written as a successful method.

The contribution is therefore a diagnostic negative result, not a renamed success claim. First, the paper records a full-matrix protocol over GQA, MMBench, TextVQA, and VQAv2, keeping full tokens, uniform pruning, FitPrune, AP-HTS, hybrid merge, and ablations in one comparable table. Second, it separates efficiency-path evidence from accuracy evidence: prefill FLOP ratios fall as visual tokens are removed, but latency and accuracy do not justify a positive frontier claim for AP-HTS. Third, it identifies the repair target: TextVQA and FitPrune together show that a spatial/activation hybrid must preserve OCR-critical tokens and must beat a simple fit-and-prune floor before it can support a positive method claim. The rest of the paper follows that evidence order: Section 2 places AP-HTS among token-reduction methods, Section 3 states the diagnostic mechanism, Section 4 defines the controlled full matrix, Section 5 reports the rejected headline, and Sections 6–7 explain why the failure concentrates in OCR-sensitive examples.

2 Related Work

Token reduction before MLLMs. Classic vision-token work studies whether image-token sequences can be shortened without losing task signal. Dynamic token sparsification, learned token selection, and token merging show that token importance can be input dependent and that merg-

ing can be preferable to hard deletion (Rao et al., 2021; Ryoo et al., 2021; Bolya et al., 2022). Earlier vision-language pretraining work also shaped the interface that pruning methods now manipulate: region features, contrastive image-text representations, and transformer fusion models all define different places where visual evidence can be lost (Anderson et al., 2018; Radford et al., 2021; Tan and Bansal, 2019; Su et al., 2020; Kim et al., 2021; Li et al., 2021). These anchors motivate AP-HTS’s merge/prune design, but they do not by themselves validate pruning inside a multimodal decoder where visual tokens interact with language queries. They also expose a design tension. Hard pruning is simple and cheap, but it can discard small regions that later become relevant to a language question. Merging preserves some aggregate signal, but it can smear text or fine-grained object evidence across a representative token. Learned token selectors can adapt to data distributions, but they introduce training cost and may not be comparable to training-free inference controls. AP-HTS was intended to occupy the training-free middle ground: keep spatial coverage, use activations as a salience signal, and test whether merging low-ranked tokens is safer than dropping them. The current evidence says that this intuition is incomplete for OCR-heavy data.

MLLM token pruning and compression.

MLLM-specific work has developed layerwise pruning, adaptive reduction, language-guided selection, packing, sparsification, and search-based retention policies (Chen et al., 2024; Shang et al., 2024; Li et al., 2024b; Ye et al., 2024; Zhang et al., 2024; Liu et al., 2024; Zhao et al., 2024; Sun et al., 2025; Yang et al., 2024; Cho et al., 2026). The same-direction ACL exemplar argues that token pruning methods can underperform simple baselines when their evaluation does not isolate the right failure modes (Wen et al., 2025). Our evidence lands in that diagnostic regime: AP-HTS is not rescued by a more elaborate salience rule once FitPrune and OCR-heavy TextVQA remain in the matrix. The FitPrune comparison is especially important because it is a strong low-overhead floor rather than a weak straw baseline. If an AP-HTS component improves one local slice but loses to FitPrune at the paper’s headline ratio, the correct conclusion is not that the component is validated; it is that the component must be integrated with a better preservation rule or replaced.

Benchmarks and evaluation. The benchmark suite intentionally mixes compositional visual reasoning, all-around multimodal assessment, OCR-oriented VQA, and standard VQA accuracy (Hudson and Manning, 2019; Liu et al., 2023b; Singh et al., 2019; Goyal et al., 2017; Antol et al., 2015). This matters because token policies that look acceptable on general visual reasoning can fail when small text regions determine the answer. The evaluation harness follows the broader lesson from multimodal evaluation work: a single retained-token ratio or one benchmark family is not enough to certify a pruning policy (Li et al., 2024a; Peng et al., 2025; Shao et al., 2025). The four selected families therefore serve different roles. GQA stresses compositional reasoning over scene structure. MMBench provides a broad multimodal assessment surface. VQAv2 checks standard visual question answering behavior. TextVQA is the stress test for small text-bearing regions and is the family where AP-HTS most clearly breaks. This diversity is why the paper does not replace the matrix with a single average score or a single hand-picked qualitative example.

3 AP-HTS Diagnostic Method

Figure 1 summarizes the evaluated mechanism and the gate that now blocks a positive claim. AP-HTS starts from a visual-token sequence and partitions image positions into a coarse spatial scaffold. The scaffold preserves coverage so that pruning does not collapse entirely onto salient central regions. Within each scaffold region, AP-HTS scores candidate tokens from observed activation magnitudes and allocates a retained-token budget across the image. Two operational variants were evaluated: direct AP-HTS pruning and a hybrid merge path that merges lower-ranked tokens into retained representatives before inference. The scaffold is meant to prevent a common failure of pure salience rules: if all retained tokens cluster around a few high-activation regions, the compressed sequence can lose spatial anchors for objects or text elsewhere in the image. The activation score is meant to prevent the opposite failure of uniform sampling: retaining coverage without regard to model state wastes budget on uninformative patches. The hybrid merge path was included because a discarded token may still carry weak evidence, and merging can in principle preserve some of that signal. The full matrix tests whether those intuitions survive contact with

real benchmark scoring.

The method was compared against full-token inference, uniform pruning, FitPrune, and AP-HTS ablations. The ablations isolate whether the scaffold, activation scoring, or pruning-only path is responsible for observed behavior. They are diagnostic rather than confirmatory: because the combined AP-HTS frontier fails, the ablations cannot support a necessity claim for the hybrid. This is important for claim calibration. If a combined method does not beat the floor, showing that one component changes the score is not enough to call that component necessary for an improved system. The FitPrune floor check is therefore part of the method definition in this evaluation. The proposed mechanism is not evaluated in isolation and then compared only in a final table; every paper-facing statement asks whether AP-HTS beats full-token drop targets, uniform pruning, and the FitPrune floor at the same retained-token ratio. This gate blocks attractive but misleading interpretations. For example, an AP-HTS win over uniform pruning on one family at ratio 0.33 is not a full-paper success when the same method loses the cross-family target and trails FitPrune at ratio 0.5.

4 Experimental Setup

Benchmarks and model. The full run evaluates GQA, MMBench, TextVQA, and VQAv2. Each benchmark-family, method, and retained-ratio cell has 128 scored tasks, producing 18,432 scored rows and 18,432 profile rows. The evaluated backend is Qwen/Qwen3-VL-8B-Instruct, with token-reduction policies applied before the language-model prefill path so that sequence length and analytical FLOP estimates reflect the compressed visual input. The task count is intentionally balanced across the four families so that the average score is not dominated by one source. Rows are paired by task id wherever a compressed condition is compared with full-token inference. The study does not claim official leaderboard performance, and it does not compare against numbers reported on different hardware or different task subsets. All reported numbers are computed from the scored benchmark examples and the released result tables.

Compared conditions. Compressed methods use retained visual-token ratios of 0.125, 0.25, 0.33, 0.5, and 0.75; the full-token reference uses ratio 1.0. The body results emphasize the 0.5 ratio because it corresponds to the target of more than 2x

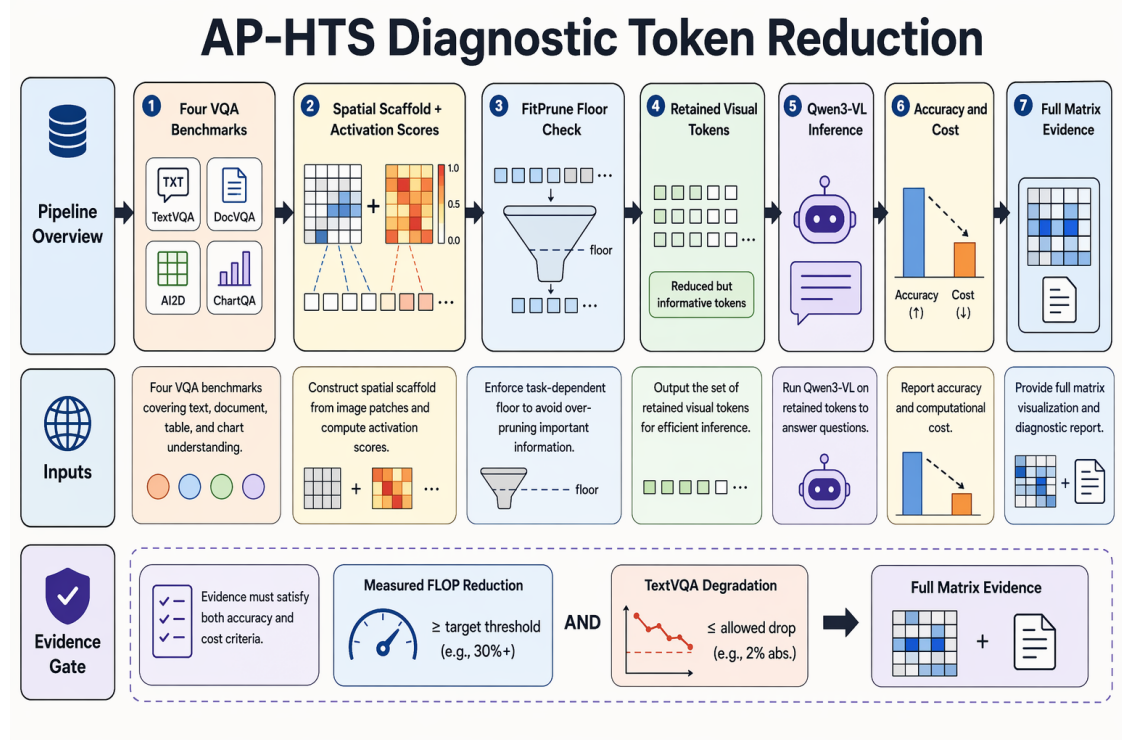


Figure 1: AP-HTS is evaluated as a diagnostic pipeline, not as a winning method: the full matrix shows measured FLOP reduction, but the FitPrune floor check blocks a positive AP-HTS frontier claim at 0.5 retained tokens.

fewer visual tokens. The compared methods are full tokens, uniform pruning, FitPrune, AP-HTS, AP-HTS hybrid merge, AP-HTS scaffold-only, AP-HTS activation-only, and AP-HTS prune-only. Uniform pruning is the sanity control: a learned or heuristic selector should beat a position-agnostic budget cut. FitPrune is the prior-floor control: a more complex diagnostic mechanism should not be considered successful if it loses to a simple training-free policy. The AP-HTS ablations are not alternate proposed methods; they are probes that help determine whether the spatial scaffold or activation signal is responsible for failures.

Metrics and uncertainty. Accuracy is computed with the frozen benchmark scorer for each task family and then averaged across families. Efficiency evidence includes measured latency rows and analytical prefill FLOP estimates computed from actual observed sequence lengths. Paired bootstrap intervals align each compressed row with its full-token counterpart by benchmark family and task id. We use these intervals as uncertainty summaries and do not claim statistical significance. This separation prevents two common overclaims. First, analytical FLOP reduction is not reported as a guaranteed wall-clock speedup, because batching, memory movement, and implementation overhead affect

measured latency. Second, a low FLOP ratio is not treated as a result if the paired accuracy drop is too large. The paper therefore reports three linked quantities—accuracy, retained-token ratio, and pre-fill FLOP ratio—but treats accuracy at matched ratio as the gate for contribution claims.

5 Main Results

Table 1 is the central body matrix. It keeps the full-token reference, the two baseline controls, and AP-HTS variants in the same view. The key result is adverse for AP-HTS: FitPrune is the only compressed method at 0.5 retained tokens that stays near the under-1% average-drop target. Full-token inference averages 0.771289 across the four benchmark families. FitPrune at ratio 0.5 averages 0.762109, close enough to the target to be the current floor. Uniform pruning averages 0.736328, showing that even the sanity control is stronger than AP-HTS hybrid merge at this ratio. AP-HTS and AP-HTS hybrid merge each average 0.692187, which is not a borderline miss but a clear rejection of the positive headline.

Figure 2 shows the same conclusion as a frontier. The AP-HTS points move left as tokens are removed, but the accuracy loss is too large to support the target claim. The frontier view also explains

Table 1: At 0.5 retained tokens, all family cells are visible: FitPrune drops only 0.009180 average accuracy while AP-HTS hybrid merge drops 0.079102.

Method	Ratio	GQA	MMB	TextVQA	VQAv2	Avg. / drop	FLOP
Full tokens	1.0	0.734	0.898	0.719	0.734	0.771 / 0.000	1.000
FitPrune	0.5	0.711	0.906	0.680	0.752	0.762 / 0.009	0.367
Uniform	0.5	0.719	0.898	0.627	0.702	0.736 / 0.035	0.367
Scaffold-only	0.5	0.719	0.898	0.627	0.702	0.736 / 0.035	0.367
Activation-only	0.5	0.688	0.859	0.419	0.667	0.658 / 0.113	0.367
Prune-only	0.5	0.711	0.867	0.451	0.709	0.685 / 0.087	0.367
AP-HTS	0.5	0.711	0.875	0.459	0.723	0.692 / 0.079	0.367
Hybrid merge	0.5	0.711	0.875	0.459	0.723	0.692 / 0.079	0.367

Each compressed row uses Qwen3-VL-8B-Instruct on 128 scored examples per family; the average/drop column is the cross-family mean accuracy and drop versus full tokens.

why cherry-picking a narrower ratio would be misleading. At very low retained ratios, every method loses substantial accuracy. At high retained ratios, AP-HTS can be less damaging, but the compression is no longer the advertised 2x setting. The body therefore keeps ratio 0.5 as the principal decision point and uses other ratios only for diagnosis.

Why the average is still useful. The cross-family average is the admission gate for a paper-level efficiency claim. The family columns then show which benchmark drives failure. Uniform pruning asks whether AP-HTS beats a task-agnostic budget cut. FitPrune asks whether it beats a stronger training-free policy at the same retained-token ratio. AP-HTS fails both tests at the headline point. It trails uniform pruning by 0.044141 average accuracy and trails FitPrune by 0.069922 average accuracy.

A conservative interpretation of local wins. Narrow local wins remain useful only as repair clues. For example, the GQA ratio-0.33 cell has a small drop, but that does not override the cross-family miss or the OCR-heavy failure. The repair question is what property helped GQA and why it did not protect TextVQA.

6 Analysis and Ablations

The ablation rows help explain what failed. Scaffold-only and activation-only variants are complete across the full matrix, but neither provides a cross-benchmark frontier that repairs the hybrid result. This weakens the original mechanism story. Spatial coverage and activation salience may each affect local behavior, yet their combination does not preserve the token types needed by the hardest family. If scaffold-only behavior resembles uniform pruning on several cells, then the scaffold

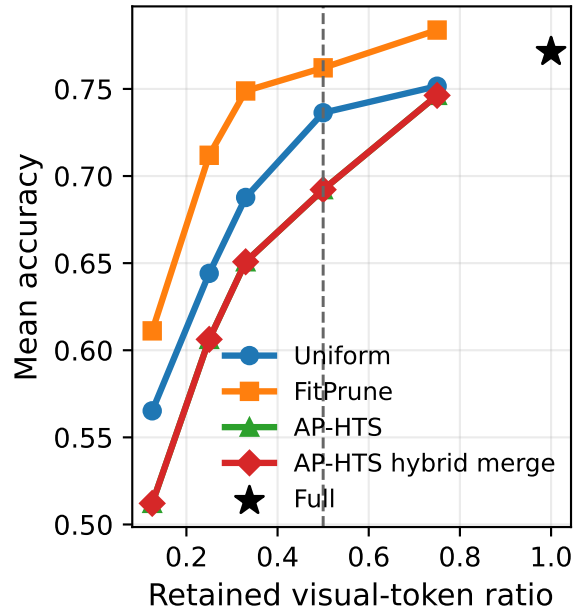


Figure 2: Accuracy-efficiency frontier: AP-HTS does not meet the 2x / under-1% target, while FitPrune is the stronger floor at the headline ratio.

mainly enforces coverage rather than semantic selectivity. If activation-only behavior improves one family but breaks another, then activation magnitude is not a sufficient proxy for question-relevant evidence. If hybrid merge tracks AP-HTS pruning at the headline point, then merging low-ranked tokens does not by itself recover the information lost by the budget rule. These findings point to a repair direction: token allocation must be conditioned on text-bearing regions or question-sensitive evidence, not only on spatial bins and activations.

What each ablation rules out. The scaffold-only condition rules out a purely geometric explanation for success. If coarse spatial coverage were enough, scaffold-only pruning would have

approached FitPrune at 0.5 retained tokens; it does not. The activation-only condition rules out a purely magnitude-based explanation. Activation magnitude reflects model state. It is not guaranteed to track the evidence that the final answer depends on, especially when the evidence is a small word or number in the image. The prune-only condition tests whether the merge operator is the source of failure. Because the hybrid merge does not repair the headline drop, the failure is better understood as a budget-allocation problem upstream of the merge operation. Together, these ablations imply that AP-HTS needs an additional preservation signal rather than a different prose framing of the same rule.

Why this remains diagnostic. The evaluated policy does not improve over FitPrune, and the related-baseline boundary still lacks a compatible FastV comparison. Those facts keep the contribution scoped to diagnosis and repair guidance.

Figure 3 separates the cost path from the accuracy claim. The run records complete latency and analytical FLOP rows for every scored condition. Lower retained ratios reduce the analytical prefill FLOP ratio, which supports the limited efficiency claim that pruning touches the expensive path. However, measured latency is not a clean proportional speedup in the current implementation. Cost reduction without acceptable accuracy is not a positive method result. This is why the efficiency evidence is limited rather than sufficient. The profile rows show that the implementation does not merely hide tokens after the expensive path has run, but they also show why actual latency must be reported alongside analytical cost.

7 Failure Cases

TextVQA is the failure surface that most clearly explains the rejected frontier. Table 2 compares paired TextVQA drops at the headline ratio. AP-HTS hybrid merge creates many more new total errors than FitPrune, consistent with a policy that removes small OCR-critical evidence even when the retained token count looks reasonable. The failure categories are paired against full-token answers. "New errors" count cases that were correct under full tokens and become completely wrong after compression. "Degraded" counts partial-score losses relative to full-token inference. For TextVQA, both categories are more damaging for AP-HTS variants than for FitPrune, which means the problem is not only a small average-score fluctu-

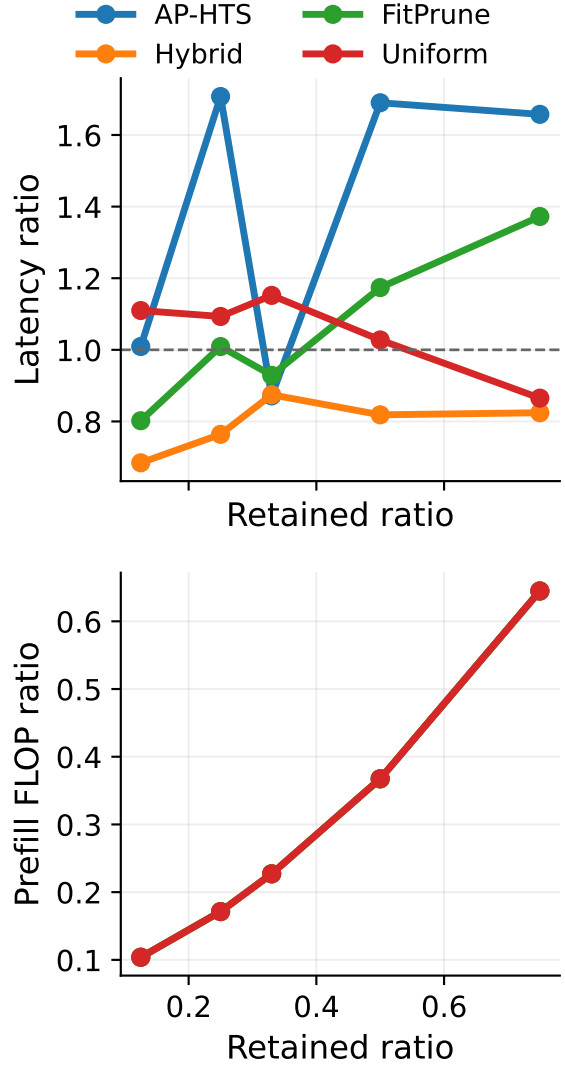


Figure 3: Efficiency evidence is real but limited: retained-token reduction lowers analytical prefill FLOPs, but does not rescue AP-HTS accuracy.

ation. The method is changing answer correctness on examples where the uncompressed model had enough evidence.

Figure 4 shows that this is not a single-row artifact. The family-level deltas reveal larger degradation in OCR-heavy conditions than in some general VQA cells. The next method-repair branch should therefore protect text-bearing regions or incorporate a query-aware OCR safeguard, then rerun the same four-family matrix against FitPrune.

Failure mechanism hypothesis. The AP-HTS scaffold protects spatial coverage at a coarse level, but TextVQA often depends on fine local evidence rather than broad coverage. A street sign, product label, handwritten number, or small document region can occupy only a few visual tokens. If those

Table 2: On TextVQA at 0.5 retained tokens, AP-HTS variants create substantially more new errors than FitPrune.

Method	Score loss		Paired changes	
	Drop	95% CI	Err.	Deg.
FitPrune	0.039	[-0.013, 0.088]	7/128	5/128
Uniform	0.092	[0.027, 0.163]	15/128	5/128
AP-HTS	0.259	[0.180, 0.338]	31/128	10/128
Hybrid	0.259	[0.173, 0.345]	31/128	10/128

tokens receive modest activation scores before the question is fully resolved, a global retained-token budget can remove them while preserving larger object regions that are visually salient but irrelevant to the answer. Merging the discarded tokens into nearby retained representatives does not guarantee recovery because OCR-like evidence is brittle: a small character sequence can be lost when mixed with background or neighboring visual texture. This hypothesis is consistent with the higher count of new TextVQA errors and with the fact that FitPrune, a simpler floor, preserves much more accuracy at the headline ratio.

Repair protocol. A credible repair run should keep the same evaluation matrix and change only the token policy. One repair path is to add an OCR-sensitive admission rule before AP-HTS allocation, so candidate tokens near text-like regions cannot be removed solely because their activation magnitude is lower than a larger object region. Another path is to make the retained-token budget query aware, allowing the question to increase the budget for image regions likely to contain text or small attributes. A third path is to use FitPrune as the base policy and add AP-HTS components only where paired diagnostics show a consistent local benefit. All three paths preserve the strongest current lesson: FitPrune is the floor, not a secondary baseline that can be hidden after the proposed method fails.

Supported claim boundaries. The evidence supports three statements: AP-HTS was evaluated on a full four-family matrix, it reduces the analytical prefill FLOP path, and it fails the target accuracy frontier. It also supports TextVQA as the critical failure surface in the current runs and FitPrune as the stronger 0.5-ratio comparator under the same experimental protocol. It does not support broad robustness, a successful AP-HTS method, or a mechanism-necessity claim from isolated table cells. These boundaries prevent an unsupported

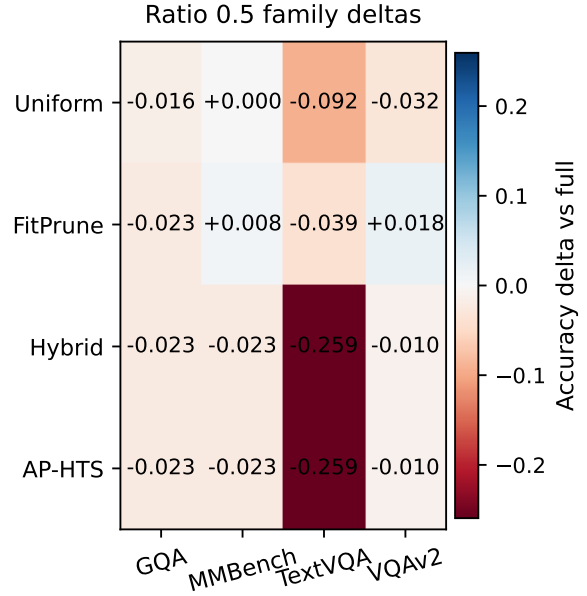


Figure 4: At 0.5 retained tokens, family-level deltas identify TextVQA as the main AP-HTS failure region under compression.

token-efficiency story.

Why the current evidence is useful. Even as a negative result, the evidence fixes the next evaluation surface: same families, ratios, metrics, uncertainty calculation, and profiling fields. A future method cannot claim progress by improving one isolated cell, hiding FitPrune, or reporting retained-token count without paired accuracy and profiling evidence.

8 Conclusion

The completed evidence does not support a positive AP-HTS frontier claim. It does support a precise repair requirement: any successor method must beat FitPrune at matched retained-token ratios and must preserve OCR-critical TextVQA evidence while retaining the measured prefill-cost reduction path. The practical acceptance criterion is therefore three-part. A repaired policy should keep the 0.5 retained-token budget as the admission point, exceed FitPrune on the cross-family average rather than only on one family, and show that lower analytical FLOPs are not bought by new OCR failures. This criterion is intentionally stricter than reporting a smaller visual-token count, because the completed matrix shows that token count, latency measurement, and answer accuracy can diverge. The current AP-HTS result fails that criterion, but it leaves a useful map: protect text-bearing evidence

first, then test any spatial or activation heuristic against the same baseline floor. The matrix also clarifies what a future positive claim must report. A repaired method should keep the family columns visible, carry the same paired full-token comparisons, and state whether TextVQA improves without hiding losses on GQA, MMBench, or VQAv2. That reporting discipline is part of the contribution because it prevents a lower token count from substituting for evidence-preserving inference.

Limitations

This study does not support a positive long-paper method claim in its current form. It evaluates one model/backend and a balanced sample from each benchmark family rather than official full benchmark test servers. FastV was not included because the available implementation did not support the evaluated Qwen3-VL backend; final comparisons require a compatible implementation or documented exclusion rationale. The balanced 128-example family slices are adequate for diagnosing the current failure mode, but they are not a substitute for official benchmark-server claims. The conceptual overview figure is explanatory only; all result claims come from benchmark result tables and paired comparisons. The cost measurements also have a bounded interpretation. Analytical prefill FLOP ratios are computed from observed sequence lengths and show that the compressed policies affect the intended expensive path, but the measured latency ratios include implementation overhead and should not be read as portable hardware speedups. For that reason, this paper treats latency and FLOPs as necessary evidence for an inference-efficiency study, not as a substitute for matched accuracy.

Ethical Considerations

The work studies inference efficiency and benchmark behavior rather than deploying a user-facing system. Lower-cost inference can make multimodal systems more accessible, but aggressive pruning can disproportionately harm text-reading and visually small evidence, which may affect accessibility and safety-critical uses. The study therefore reports the negative TextVQA finding directly and does not recommend AP-HTS for deployment. The same failure mode is relevant beyond the benchmark setting. Applications that rely on signs, labels, or small visual details should not adopt token-reduction policies based only on aggregated accuracy or retained-token counts. Any deployment-oriented variant would need targeted evaluation on text-heavy and safety-relevant images before claiming reduced-cost inference as a benefit. Reporting the failed frontier is therefore also an ethical choice: it prevents an efficiency story from masking predictable errors on small or written visual evidence.

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A Reproducibility Notes

The canonical evidence source is the completed full matrix with 18,432 scored rows and 18,432 profile rows. The reproducibility bundle includes the scored result table, profile summaries, analysis tables, data figures, and accepted conceptual overview figure. The evaluation uses fixed task ids within each benchmark family and paired comparisons against full-token rows. It also records task